ModelIt!

Teacher's Guide

Can We Build a Fusion Reactor?

Modeling Plasma Confinement, Energy Balance, and the Path to Unlimited Clean Energy

Quick Reference

Grade Level	Grade 11L3
NGSS Standard	HS-PS1-8, HS-PS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS1-8, HS-PS3-2

Develop models to illustrate the changes in the composition of the nucleus of the atom and the energy released during the processes of fission, fusion, and radioactive decay. Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motions of particles and energy associated with the relative positions of particles.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to illustrate how the interplay between plasma physics parameters determines whether a fusion reactor achieves net energy gain or remains an energy consumer.
Disciplinary Core Idea	**PS1.C Nuclear Processes / PS3.B Conservation of Energy and Energy Transfer** — Students model how nuclear fusion converts mass to energy via E=mc squared and trace energy flows from nuclear binding energy through plasma heating, neutron capture, and electrical conversion.
Crosscutting Concept	**Energy and Matter** — Students analyze how matter transforms from deuterium-tritium fuel to helium ash while energy flows from nuclear potential energy through kinetic, thermal, and electrical forms, quantifying conversion efficiencies at each stage.

CAST Testing Alignment

MULTIPLE CHOICE:

A fusion reactor achieves a plasma temperature of 150 million degrees and adequate fuel density, but magnetic confinement fluctuates causing confinement time to drop by 40%. According to the Lawson criterion and your model, what is the most likely consequence for net energy gain?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why achieving Q greater than 1 was a historic milestone but Q greater than 10 is required for commercial power. Describe the energy inputs that Q must overcome and why each one is necessary for the reactor to function.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every star in the universe is a fusion reactor, converting hydrogen into helium and releasing staggering amounts of energy. Our Sun alone converts 600 million tons of hydrogen per second, a process that has been running continuously for 4.6 billion years and will continue for 5 billion more. Recreating this process on Earth would provide virtually unlimited clean energy — enough to power civilization for billions of years using fuel extracted from seawater. The physics has been understood since the 1920s. The engineering has defied humanity's best efforts since the 1950s.

The Physics of Fusion

Nuclear fusion occurs when two light atomic nuclei overcome their electromagnetic repulsion and merge, releasing energy from the strong nuclear force. The most achievable reaction fuses deuterium (hydrogen with one extra neutron) with tritium (hydrogen with two extra neutrons) to produce a helium-4 nucleus, a 14.1 MeV neutron, and 17.6 MeV of total energy. This is the same process that powers the Sun, but the Sun cheats: its enormous gravitational pressure compresses hydrogen to extreme density, allowing fusion at a mere 15 million degrees. Without gravity's assistance, Earth-based reactors must reach 150 million degrees — ten times hotter than the Sun's core — to achieve useful fusion rates.

The Confinement Challenge

At 150 million degrees, matter exists as plasma — a sea of charged particles that would vaporize any physical container on contact. Magnetic confinement uses powerful superconducting magnets to create a magnetic cage that suspends the plasma in vacuum. The most successful design is the tokamak, a donut-shaped chamber where spiraling magnetic field lines confine the plasma. But plasma is inherently unstable: magnetohydrodynamic instabilities, edge-localized modes, and turbulent transport constantly push plasma toward the walls. Maintaining stable confinement requires real-time feedback control systems adjusting magnetic fields thousands of times per second — the engineering equivalent of balancing an inverted pendulum on a moving platform.

The Materials Crisis

Each deuterium-tritium fusion reaction produces a 14.1 MeV neutron that carries 80% of the reaction's energy. Unlike charged particles, neutrons are immune to magnetic confinement and slam directly into the reactor's structural materials. Over years of operation, these neutrons displace atoms from crystal lattices, create helium bubbles that swell and embrittle steel, and transmute structural elements into different atoms. No existing material can withstand the neutron flux of a commercial fusion reactor for its intended 30-year lifespan. Reduced-activation ferritic-martensitic steels, tungsten alloys, and silicon carbide composites are leading candidates, but all eventually succumb to neutron damage.

The Race to Commercial Fusion

In December 2022, the National Ignition Facility achieved scientific breakeven (Q greater than 1) using inertial confinement. ITER, the 35-nation international tokamak in southern France, aims to achieve Q=10 by the early 2030s — producing ten times more energy than it consumes. Meanwhile, private fusion companies have raised over $6 billion: Commonwealth Fusion Systems is building SPARC, a compact tokamak using high-temperature superconducting magnets; Helion Energy is pursuing field-reversed configuration for direct electricity generation; and TAE Technologies uses beam-driven field-reversed configuration. The consensus that fusion is perpetually 30 years away is shifting — multiple credible pathways now aim for demonstration plants in the 2030s.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven key components of a fusion reactor system: plasma temperature, magnetic field strength, fuel density, energy output, neutron flux, wall material degradation, and net energy gain.

E **Establish Relationships**

Students map relationships showing that the three input parameters (temperature, field strength, density) must simultaneously satisfy the Lawson criterion, while increased fusion output inherently increases neutron damage — revealing the fundamental engineering tension.

V **Visualize & Model**

Students build a computational model showing the seven-component fusion system with the Lawson threshold, energy balance calculations, and material degradation rates.

E **Evaluate Outcomes**

Students run ITER design parameters, push-to-ignition, and confinement collapse scenarios to identify the operating envelope where net energy gain is achievable without unacceptable material damage.

R **Revise & Extend**

Students add tritium breeding ratio, plasma disruption frequency, or divertor heat load to model more complete fusion reactor physics and engineering constraints.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The Sun does this effortlessly — 600 million tons of hydrogen fused every second for almost 5 billion years. We have spent 70 years and over 50 billion dollars trying to do it in a room. Today you find out why it is so incredibly hard."*

Watch For

Open with the contrast: the Sun's effortless fusion versus humanity's decades of struggle. Show the scale of ITER under construction. Ask: what does the Sun have that we do not?

Slide 2: Learning Objectives

3 min

Say: *"You are about to model the most extreme engineering challenge on Earth — a device that must simultaneously be ten times hotter than the Sun's core and colder than outer space, with walls that survive decades of neutron bombardment."*

Watch For

Have students read objectives. Pre-teach 'Lawson criterion' as the three-dimensional threshold that defines success or failure. Pre-teach 'Q factor' as the scoreboard.

Slide 3: Big Question

3 min

Say: *"The answer is not that the physics does not work — every star proves it works. The answer is that ENGINEERING it on Earth requires solving problems at simultaneous extremes that have never coexisted in any machine humans have built."*

Watch For

Quick-write: Students list what they think the biggest engineering challenges are. Share out. Track guesses on the board — revisit at the end.

Slide 4: LEVER Framework

2 min

Say: *"We are modeling a system where three inputs must all cross a critical threshold simultaneously, and the rewards and the damage come from the exact same nuclear reactions."*

Watch For

Preview LEVER steps. Emphasize the Lawson criterion as a three-dimensional boundary and the output-damage coupling as the fundamental trade-off.

Slide 5: Activity 1: Components

8 min

Say: *"Seven components. Three inputs you control: temperature, magnetic field, fuel density. Four outputs that physics dictates. The catch: the outputs you want and the outputs you do not want are inseparable products of the same reaction."*

Watch For

Guide students through all components. Discuss why the three inputs are external and the four outputs are internal. Emphasize the 80-20 split of fusion energy between neutrons and alpha particles.

Slide 6: Activity 2: Connections

8 min

Say: *"Follow the energy: temperature plus density plus confinement time must cross the Lawson threshold. When they do, you get fusion energy AND neutron bombardment. You cannot have one without the other."*

Watch For

Help students map the coupled system. Use the Lawson criterion as an organizing framework. Highlight the cruel paradox: more fusion output always means more wall damage.

Slide 7: Activity 3: Simulation

10 min

Say: *"Can we reach Q=10? What happens when we push harder? And what does a confinement failure look like? Let us find out."*

Watch For

Students predict Q factor and damage rates before running each scenario. The confinement collapse is the key insight — watch how quickly everything unravels when the magnetic bottle weakens.

Slide 8: Discoveries

5 min

Say: *"Now you understand the real reason fusion has been thirty years away for sixty years. It is not one problem — it is a system of coupled problems where solving one often makes another harder."*

Watch For

Connect model findings to real-world progress. NIF ignition, ITER construction, private company breakthroughs. Ask: is the thirty-year curse finally breaking?

Slide 9: STEM Challenge

12 min

Say: *"The DOE wants your ideas. Pick the biggest bottleneck and propose a breakthrough. The energy future of humanity might depend on your generation solving this."*

Watch For

Groups identify their chosen bottleneck and design innovative approaches. Must reference model data. Present and defend. Class evaluates feasibility and creativity.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Plasma Temperature, Magnetic Field Strength, Fuel Density
- **INTERNAL:** Energy Output, Neutron Flux, Wall Material Degradation, Net Energy Gain

Relationships

- **Plasma Temperature to Energy Output:** POSITIVE (+) — Higher plasma temperature increases the kinetic energy of deuterium and tritium nuclei, dramatically increasing the probability that they overcome Coulomb repulsion and fuse. The fusion cross-section peaks at around 150 million degrees, so higher temperature means exponentially more fusion reactions and greater energy output.
- **Energy Output to Neutron Flux:** POSITIVE (+) — Higher energy output means more deuterium-tritium fusion reactions per second, and every single reaction produces exactly one 14.1 MeV neutron. The neutron flux is directly proportional to the fusion reaction rate — you cannot produce fusion energy without producing neutrons.
- **Neutron Flux to Wall Material Degradation:** POSITIVE (+) — Higher neutron flux means more high-energy neutrons bombarding the reactor first wall every second, displacing more atoms from their crystal lattice positions, creating more helium bubbles, and transmuting more structural atoms. Material degradation rate is directly proportional to neutron flux integrated over time.
- **Magnetic Field Strength to Energy Output:** POSITIVE (+) — Stronger magnetic fields confine plasma more tightly, increasing fuel density and extending confinement time — both of which increase the fusion reaction rate and energy output by helping satisfy the Lawson criterion.

Key Discoveries

- The Lawson criterion creates a three-dimensional threshold problem — falling short on any single parameter (temperature, density, or confinement time) means zero net energy, even if the other two are perfect
- Every fusion reaction that produces useful energy also produces a high-energy neutron that damages the reactor, creating a fundamental trade-off between energy output and structural lifetime
- Plasma instabilities are the primary obstacle to sustained confinement — turbulence, edge-localized modes, and disruptions constantly try to breach the magnetic bottle, requiring real-time feedback control thousands of times per second
- The enormous energy cost of heating plasma, powering magnets, and running cooling systems means that the fusion reactions must produce not just some energy but enough to overcome all parasitic loads with margin

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Fusion reactors can explode like nuclear bombs"	A fusion reactor cannot produce a nuclear explosion. Nuclear weapons require precisely engineered implosion to compress fusion fuel to extreme density in nanoseconds — conditions that cannot occur accidentally. A fusion reactor contains only about one gram of fuel at any moment. If confinement fails, the plasma cools to room temperature in milliseconds and fusion stops. A reactor failure damages equipment — potentially costing billions to repair — but does not produce a nuclear explosion. The physics of weapons and reactors are fundamentally different despite both involving fusion.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We already know how to build a fusion reactor, we just need more funding"	While the basic physics of fusion is well understood, several critical engineering challenges remain unsolved. No material has been demonstrated to survive a commercial reactor's neutron flux for 30 years. Tritium self-sufficiency (breeding ratio greater than 1) has never been achieved. Steady-state plasma confinement for months or years (versus seconds in current experiments) has not been demonstrated. And the economics of fusion electricity versus increasingly cheap renewables is uncertain. More funding would accelerate progress, but money alone cannot solve problems that require fundamental materials science and engineering breakthroughs.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Fusion produces no waste and is perfectly clean"	Fusion is dramatically cleaner than fission or fossil fuels, but it is not waste-free. The 14.1 MeV neutrons from fusion activate structural materials, creating low-level radioactive waste that must be stored for 50-100 years before it decays to safe levels. Tritium is radioactive with a 12.3-year half-life and must be carefully contained. However, fusion produces no long-lived actinides (unlike fission's plutonium that remains dangerous for hundreds of thousands of years), no greenhouse gases, and no possibility of meltdown. The waste comparison: 50-100 years of storage versus 100,000+ years is an enormous improvement.